

Yifei Wang

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Ph.D. in Computer Science and 1 year industrial postdoctoral experience specializing in Deep Learning and AI for small molecule design. Architected and deployed production-level molecular design workflows, including efficient virtual screening tools and synthesizable generative AI tools. Translated medicinal chemistry constraints into robust, specialized AI algorithms through a proven track record of cross-functional collaboration with computational chemists, effectively bridging the gap between advanced ML research and practical drug discovery applications.

RESEARCH TOPICS

Machine Learning, Deep Learning, Efficient Search, Optimization, Graph Neural Network, Generative AI, Reinforcement Learning, Transformer, Chemical Language Model, AI For Drug Discovery.

WORK EXPERIENCE

- Biogen** Feb 2025 - Present
Machine Learning Research Fellow Massachusetts, USA
 - Engineered and deployed a Thompson Sampling-based search tool for ligand-based virtual screening, enabling billion-level screening in 1 day with demonstrated coverage across internal small-molecule projects.
 - Developed a synthesizable molecular generation tool for multi-step *de novo* molecular design, integrating deep generative models and reinforcement learning within iterative "generate → score → learn" design loops.
- Brigham and Women's Hospital** Aug 2022 - Jan 2025
Research Trainee part-time Massachusetts, USA
 - Improved reliability and interpretability of healthcare AI models using EHR data across ML and LLMs.
 - Detected and evaluated fairness and bias in healthcare AI decision systems.
- XtalPi Inc.** Jul 2019 - Dec 2019
Algorithm Intern for AI-aided drug discovery Shenzhen, China
 - Implemented a transformer-like architecture for retrosynthetic reaction path prediction.
 - Improved model architectures and training strategies. Refined the pipeline of reaction prediction.

EDUCATION

- Brandeis University** Jan 2020 - Feb 2025
Ph.D. in Computer Science Massachusetts, USA
 - Research Focus: Deep Learning, Graph Neural Network, Molecular Learning, Generative AI, Language Model.
- Tsinghua University** Aug 2015 - Jul 2019
Bachelor of Pure and Applied Mathematics Beijing, China
 - Major Focus: Statistics, Efficient Algorithm Design, Optimization, Machine Learning.

SELECTED PUBLICATIONS

- Yifei Wang***, Yunrui Li*, Pengyu Hong, Hao Xu. "Advancing Drug Discovery with Enhanced Chemical Understanding via Asymmetric Contrastive Multimodal Learning." *Journal of Chemical Information and Modeling*, 2025.
- Yifei Wang**, Liqin Wang, Zhengyang Zhou, John Laurentiev, Joshua R. Lakin, Li Zhou, and Pengyu Hong. "Assessing fairness in machine learning models: A study of racial bias using matched counterparts in mortality prediction for patients with chronic diseases." *Journal of Biomedical Informatics*, 2024.
- Yifei Wang***, Zhengyang Zhou*, Liqin Wang, John Laurentiev, Peter Hou, Li Zhou, Pengyu Hong. "Counterpart Fairness – Addressing Systematic Between-group Differences in Fairness Evaluation" In *NeurIPS Workshop*, 2024.
- Yifei Wang**, Shiyang Chen, Guobin Chen, Ethan Shurberg, Hang Liu, and Pengyu Hong. "Motif-Based Graph Representation Learning with Application to Chemical Molecules." *Informatics*, vol. 10, no. 1, p. 8. MDPI, 2023.

- [5] Tong Yang*, Yifei Wang*, Long Sha*, Jan Engelbrecht, and Pengyu Hong. "Knowledgebra: An Algebraic Learning Framework for Knowledge Graph." *Machine Learning and Knowledge Extraction* 4, no. 2 (2022): 432-445.
- [6] Li, Peizhao, Yifei Wang, Han Zhao, Pengyu Hong, and Hongfu Liu. "On dyadic fairness: Exploring and mitigating bias in graph connections." In *International Conference on Learning Representations (ICLR)*, 2021.

SELECTED PROJECTS

• Project 1: Reinforcement-Learning-Driven Synthesizable Molecular Generative Design

- Developed a Transformer-based generative method for synthesizable molecular design that produces molecules alongside their corresponding synthetic pathways.
- Implemented goal-directed generation using on-policy reinforcement learning (PPO) to improve molecular structures while navigating multi-billion-scale chemical spaces.
- Adapted the model for specialized molecular design requirements, incorporating constraints to protect privileged groups and preferred reactions.

• Project 2: Motif-based Graph Representation Learning on Molecules

[\[Paper\]](#)[\[Code\]](#)

- Designed a novel convolution module for graph representational learning on molecules with an efficient pretraining strategy, enabling the capture of local structural and semantic information from graph motifs.
- Demonstrated the superiority of the proposed model in molecular learning across multiple dimensions: (1) better results in various molecular property prediction tasks, (2) more stable training, and (3) improved interpretability.

• Project 3: Explainable Contrastive Multimodal Learning on Molecules

[\[Paper\]](#)[\[Code\]](#)

- Proposed a training-efficient framework promoting cross-modality understanding between 2D molecular graphs and other chemical modalities.
- Demonstrated the effectiveness and interpretability of the proposed framework in various challenging tasks: (1) large-scale cross-modality retrieval, (2) isomer discrimination, and (3) ADMET prediction.

SKILLS

- **ML Frameworks:** PyTorch, Pytorch Geometric, Huggingface, Transformer Reinforcement Learning, Chemprop
- **Molecular Design:** Deep Reinforcement Learning (RL), Generative Models (Diffusion, GNNs, Transformers), Efficient Search (Thompson Sampling, Combinatorial Search).
- **Cheminformatics:** RDKit, SMILES/SMARTS Mapping, Virtual Screening.
- **Soft Skills:** Project and Time Management, Collaboration, Scientific Paper Writing, Mentorship.

SERVICES

- **Reviewer:** NeurIPS, ICLR, ICML, COLING, KDD Workshop, NIPS AI4Science Workshop, AMIA Annual Symposium, International Journal of Medical Informatics, BMC Medical Informatics and Decision Making
- **Teaching Assistant:** Computation Theory, Spring 2024 & Statistical Machine Learning, Fall 2020, 2021, 2022.
- **Undergraduate Thesis Advisor:** Supervised a thesis that earned the Kukin Moskowitz Prize for Excellence 2024.

*Equal contribution.